Polygon conjoinability: definitions, proofs, and a formal version of the tension and resonance hypotheses

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Abstract

Take a regular n-gon P and conjoin to it copies of itself, each copy being the reflection of P across one of its edges, so that no two copies overlap. The *conjoinability* $k(n)$ of the n-gon is the largest number of copies that can be conjoined in this way. We prove that $k(n) = \lfloor n/\lceil n/6 \rceil \rfloor$ and compute its values explicitly, together with the *slack* $\sigma(n) = n - k(n)\lceil n/6 \rceil$ and the number $s(n) = \max(0, \lceil n/6 \rceil - 2)$ of central vertices left untouched between two adjacent copies at the tightest packing. In particular the conjoinability equals 6 exactly when $6 \mid n$. The proof reduces overlap of two copies to the membership of their centre offset in a difference body, which gives the exact threshold $\lceil n/6 \rceil$. We then count all valid placements (exactly, up to rotation, and up to rotation and reflection), characterise the n for which the extremal placement is unique, identify the placement polynomial with a power sum of the roots of $\lambda^m = \lambda^{m-1} + x$, define the *tension* $\tau(n)$ of the tightest packing, prove that it vanishes exactly at the hexagonal resonances, and bound its distance from an explicit sawtooth profile for large n by an explicit inequality. All statements are cross-checked against an exact-geometry program (Section 9).

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1 Setting and definitions

1.1 The polygon and its edges

Fix an integer $n \geq 3$ and put $u := \pi/n$. Let P be the regular n-gon with centre at the origin O and circumradius 1 (the scale plays no role). Its vertices are $v_j = (\cos 2ju,\ \sin 2ju)$ for $j \in \mathbb{Z}_n$. Edge $e \in \mathbb{Z}_n$ is the segment from v_e to v_{e+1}. Its midpoint $\frac{1}{2}(v_e + v_{e+1})$ lies at angle $\psi_e := (2e+1)u$ and at distance

$$a := \cos u \qquad \text{(the apothem)}$$

from O, and the side length is

$$\ell := |v_{e+1} - v_e| = 2\sin u.$$

The outward unit normal of edge e is $u_e := (\cos\psi_e, \sin\psi_e)$ (not to be confused with the number $u = \pi/n$; we always write the normal with its index). The line carrying edge e is $L_e := \{x \in \mathbb{R}^2 : \langle x, u_e\rangle = a\}$, and $P \subseteq \{x : \langle x, u_e\rangle \leq a\}$.

1.2 Neighbours, placements, conjoinability

Definition 1.1 (Neighbour)**.** Let $\mathcal{R}_e$ be the reflection in the line L_e, that is

$$\mathcal{R}_e(x) = x - 2\big(\langle x, u_e\rangle - a\big)\, u_e.$$

The *neighbour on edge* e is $N_e := \mathcal{R}_e(P)$. We say that N_e is *conjoined* to P along edge e.

Since $\langle x, u_e\rangle - a \leq 0$ for $x \in P$ and $\mathcal{R}_e$ changes the sign of $\langle x, u_e\rangle - a$, we have $N_e \subseteq \{x : \langle x, u_e\rangle \geq a\}$. So N_e lies on the other side of L_e and its interior is disjoint from P; the polygons P and N_e share exactly the edge e.

Definition 1.2 (Placement)**.** A set $S \subseteq \mathbb{Z}_n$ of edges is a *placement* if $\operatorname{int} N_e \cap \operatorname{int} N_f = \emptyset$ for all distinct $e, f \in S$. The *size* of S is $|S|$. Touching (along a point or a segment) is allowed; overlapping on top of another copy is not.

Definition 1.3 (Conjoinability)**.** The *conjoinability* of the regular n-gon is

$$k(n) := \max\{\, |S| :\ S \subseteq \mathbb{Z}_n \text{ is a placement}\,\},$$

the largest number of copies of P that can be conjoined to P along full edges by reflection without overlap. A placement of size $k(n)$ is called *extremal*.

1.3 Spacing, vertex skipping, slack and tension

The *cyclic distance* of two distinct edges is $d(e, f) = \min\big(|e-f|,\, n - |e-f|\big) \in \{1, \ldots, \lfloor n/2\rfloor\}$, where $|e-f|$ is computed from representatives in $\{0, \ldots, n-1\}$. The rotation of the plane by $2\pi/n$ maps P to itself, v_j to v_{j+1} and N_e to N_{e+1}. Hence whether N_e and N_f overlap depends only on $d(e, f)$.

Definition 1.4 (Minimal spacing)**.** A distance $d \geq 1$ is *admissible* if $\operatorname{int} N_e \cap \operatorname{int} N_f = \emptyset$ whenever $d(e, f) = d$. The *minimal spacing* $m(n)$ is the smallest admissible distance. (It exists, and equals $\lceil n/6\rceil$, by Theorem 3.2 and Proposition 3.4.)

For a placement $S = \{e_1 < e_2 < \cdots < e_t\}$ with $t \geq 2$ the *gaps* are $g_i = e_{i+1} - e_i$ for $1 \leq i < t$ and $g_t = n - e_t + e_1$; they satisfy $g_1 + \cdots + g_t = n$.

Definition 1.5 (Vertex skipping and slack)**.** Two neighbours on the edges e and $e+m$, $m = m(n)$, leave the central vertices $v_{e+2}, \ldots, v_{e+m-1}$ untouched, which are $m-2$ vertices when $m \geq 2$. The *vertex skipping* is $s(n) := \max(0,\, m(n) - 2)$ (for $m = 1$ the neighbours share the vertex v_{e+1} and nothing is skipped). The *slack* is $\sigma(n) := n - k(n)\, m(n)$, the number of edges of P left over when $k(n)$ neighbours are packed at the minimal spacing.

Definition 1.6 (Tension)**.** Let e, f be edges with $d(e,f) = m(n)$. The *tension* of the n-gon is the clearance between the two neighbours, measured in sides of the polygon:

$$\tau(n) := \frac{\operatorname{dist}(N_e, N_f)}{\ell}, \qquad \ell = 2\sin\tfrac{\pi}{n}.$$

By the rotation symmetry above it does not depend on the choice of e, f. Always $\tau(n) \geq 0$. We call the packing *maximally tense* if $\tau(n) = 0$, i.e. if the neighbours press against each other, and *loose* if $\tau(n) > 0$. Tension is a geometric quantity (a length); the slack $\sigma(n)$ is its combinatorial counterpart (a number of edges).

Symbol	Meaning
$u = \pi/n$	half the central angle subtended by one edge
$a = \cos u$	apothem of P (distance from O to an edge)
$\ell = 2\sin u$	side length of P
$u_e,\ \psi_e,\ L_e,\ \mathcal{R}_e$	outward unit normal of edge e, its angle $(2e+1)u$, the line carrying the edge, the reflection in that line
$N_e = \mathcal{R}_e(P)$	neighbour on edge e
$d(e,f)$	cyclic distance of two edges
$m(n)$	minimal admissible spacing ($= \lceil n/6 \rceil$)
$k(n)$	conjoinability: largest number of non-overlapping neighbours
$\sigma(n),\ s(n)$	slack $n - k\,m$; vertex skipping $\max(0, m-2)$
$\tau(n)$	tension: clearance of the tightest packing in sides
$D = P - P$	difference body $\{p - q :\ p, q \in P\}$
$h_D,\ r_D,\ R_D$	support function, inradius and circumradius of D
$c_{ef} = 2a(u_f - u_e)$	offset between the centres of N_e and N_f
$q,\ r,\ \kappa$	$n = 6q + r$ with $q = \lfloor n/6 \rfloor$, $r = n \bmod 6$; $\kappa = (6-r)/6$
$N(n,t),\ P_n(x)$	number of placements of size t; placement polynomial

Table 1: Notation.

2 The geometry of neighbours

Lemma 2.1 (Neighbours are translates of $-P$)**.** *$N_e = 2a\,u_e + (-P)$ for every edge e.*

Proof. From Definition 1.1, $\mathcal{R}_e(x) = \big(x - 2\langle x, u_e\rangle u_e\big) + 2a\,u_e$. Put $T_e(x) := x - 2\langle x, u_e\rangle u_e$, the reflection in the line through O perpendicular to u_e, and $M_e(x) := 2\langle x, u_e\rangle u_e - x$, the reflection in the line Ou_e. Then $T_e(x) = -M_e(x)$ (expand: $-M_e(x) = x - 2\langle x, u_e\rangle u_e$). The line Ou_e passes through the centre of P and the midpoint of edge e, which is a symmetry axis of the regular polygon, so $M_e(P) = P$. Therefore

$$N_e = \mathcal{R}_e(P) = 2a\,u_e + T_e(P) = 2a\,u_e - M_e(P) = 2a\,u_e - P. \qquad \square$$

Lemma 2.2 (Translates of a convex body)**.** *Let $K \subset \mathbb{R}^2$ be a convex body (compact, convex, non-empty interior) and $x, y \in \mathbb{R}^2$. Then*

(a) $(x+K) \cap (y+K) \neq \emptyset \iff y - x \in K - K$*;*

(b) $\operatorname{int}(x+K) \cap \operatorname{int}(y+K) \neq \emptyset \iff y - x \in \operatorname{int}(K-K)$*;*

(c) $\operatorname{dist}(x+K,\, y+K) = \operatorname{dist}(y-x,\, K-K)$*,*

where $K - K = \{p - q :\ p, q \in K\}$.

Proof. (a) If $z \in (x+K) \cap (y+K)$ then $z - x \in K$ and $z - y \in K$, so $y - x = (z-x) - (z-y) \in K - K$. Conversely, if $y - x = p - q$ with $p, q \in K$, then $z := x + p = y + q$ lies in both sets.

(b) For convex sets C_1, C_2 the relative interiors satisfy $\operatorname{ri}(C_1 + C_2) = \operatorname{ri} C_1 + \operatorname{ri} C_2$ (Rockafellar, *Convex Analysis*, Cor. 6.6.2). With $C_1 = K$, $C_2 = -K$ this gives $\operatorname{int}(K-K) = \operatorname{int} K - \operatorname{int} K$. If

$z \in \operatorname{int}(x+K) \cap \operatorname{int}(y+K)$, then $z-x$, $z-y \in \operatorname{int} K$ and $y-x = (z-x)-(z-y) \in \operatorname{int} K - \operatorname{int} K = \operatorname{int}(K-K)$. Conversely, if $y-x = p-q$ with $p, q \in \operatorname{int} K$, then $z := x+p = y+q$ lies in both interiors.

(c) $\operatorname{dist}(x+K, y+K) = \inf_{p,q\in K} \big|(x+p)-(y+q)\big| = \inf_{p,q\in K} \big|(x-y)+(p-q)\big| = \operatorname{dist}(x-y, -(K-K)) = \operatorname{dist}(y-x, K-K)$, because $K-K$ is symmetric about the origin. □

Put

$$D := P - P = P + (-P),$$

a centrally symmetric convex body with 0 in its interior. For two edges e, f with $d(e,f) = d$ put $c_{ef} := 2a(u_f - u_e)$.

The length of c_{ef}. Since $|u_f - u_e|^2 = 2 - 2\cos(\psi_f - \psi_e) = 4\sin^2 \frac{\psi_f - \psi_e}{2}$ and $\psi_f - \psi_e = 2u(f-e)$, we get $|u_f - u_e| = 2\,|\sin(u(f-e))| = 2\sin(ud)$ (the values for $f-e$ and $n-(f-e)$ agree because $\sin(\pi-\theta) = \sin\theta$). Hence

$$|c_{ef}| = 4a\sin(ud) = 4\cos u \,\sin\tfrac{\pi d}{n}. \tag{1}$$

Corollary 2.3 (Overlap criterion)**.** $\operatorname{int} N_e \cap \operatorname{int} N_f \neq \emptyset \iff c_{ef} \in \operatorname{int} D$, *and* $\operatorname{dist}(N_e, N_f) = \operatorname{dist}(c_{ef}, D)$.

Proof. By Lemma 2.1, $N_e = O_e + K$ with $K = -P$ and $O_e = 2a\,u_e$, and $K-K = (-P)-(-P) = -P+P = D$. Apply Lemma 2.2 with $x = O_e$, $y = O_f$, so $y - x = O_f - O_e = c_{ef}$. □

Lemma 2.4 (Shape of D)**.** *Let $h_D(\theta) = \max_{x\in D}\langle x, (\cos\theta, \sin\theta)\rangle$ be the support function of D, and put $r_D := \min_\theta h_D(\theta)$, $R_D := \max_\theta h_D(\theta)$. Then $\{|x| \le r_D\} \subseteq D \subseteq \{|x| \le R_D\}$, and*

$$\begin{aligned} &n \text{ even}: && r_D = 2a = 2\cos u, && R_D = 2,\\ &n \text{ odd}: && r_D = 1 + a = 1 + \cos u = 2\cos^2\tfrac{u}{2}, && R_D = 2\cos\tfrac{u}{2}. \end{aligned}$$

Proof. Write $w_\theta = (\cos\theta, \sin\theta)$. A convex body is the intersection of its supporting half planes, $D = \bigcap_\theta \{x : \langle x, w_\theta\rangle \le h_D(\theta)\}$; every half plane contains the disc of radius r_D, so that disc lies in D. For $x \in D$, $x \neq 0$, we have $|x| = \langle x, x/|x|\rangle \le h_D(x/|x|) \le R_D$.

Support function of P. $h_P(\theta) = \max_j \langle v_j, w_\theta\rangle = \max_j \cos(\theta - 2ju) = \cos\delta(\theta)$, where $\delta(\theta) := \operatorname{dist}(\theta, 2u\mathbb{Z}) \in [0, u]$ is the angular distance to the nearest vertex direction (cosine is decreasing on $[0,\pi]$). Since $D = P + (-P)$ and $h_{-P}(\theta) = h_P(\theta+\pi)$, we get $h_D(\theta) = h_P(\theta) + h_P(\theta+\pi) = \cos\delta(\theta) + \cos\delta(\theta+\pi)$.

n even. Then $\pi = \frac{n}{2}\cdot 2u \in 2u\mathbb{Z}$, so $\delta(\theta+\pi) = \delta(\theta)$ and $h_D = 2\cos\delta$. As δ ranges over $[0,u]$, the minimum is $2\cos u$ and the maximum is 2.

n odd. Then $\pi = \frac{n-1}{2}\cdot 2u + u$, so $\delta(\theta+\pi) = \operatorname{dist}(\theta+u, 2u\mathbb{Z})$. If $\theta \equiv +\delta$ modulo $2u$, then $\theta + u \equiv u + \delta \in [u, 2u]$ and its distance to $2u\mathbb{Z}$ is $2u - (u+\delta) = u - \delta$. If $\theta \equiv -\delta$, then $\theta + u \equiv u - \delta \in [0, u]$, with distance $u - \delta$. So $\delta(\theta+\pi) = u - \delta(\theta)$ in both cases and, by the sum-to-product formula $\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$ with $A = \delta$, $B = u - \delta$,

$$h_D = \cos\delta + \cos(u-\delta) = 2\cos\tfrac{u}{2}\,\cos\big(\delta - \tfrac{u}{2}\big), \qquad |\delta - \tfrac{u}{2}| \le \tfrac{u}{2}.$$

The maximum is $2\cos\frac{u}{2}$ (at $\delta = \frac{u}{2}$) and the minimum is $2\cos\frac{u}{2}\cos\frac{u}{2} = 2\cos^2\frac{u}{2} = 1 + \cos u$ (at $\delta \in \{0, u\}$). □

3 The spacing theorem

Lemma 3.1 (Two trigonometric inequalities)**.** *Let $S(u) = \sin\big(\frac{\pi}{6} + \frac{7u}{6}\big) + \sin\big(\frac{\pi}{6} - \frac{5u}{6}\big)$. Then*

(i) $S(u) > 1$ for $0 < u \le \pi/8$;

(ii) $S(u) > \cos\frac{u}{2}$ for $0 < u \le \pi/7$.

Proof. Differentiating twice (each argument has derivative $\frac{7}{6}$ resp. $-\frac{5}{6}$),

$$S''(u) = -\tfrac{49}{36}\sin\!\big(\tfrac{\pi}{6}+\tfrac{7u}{6}\big) - \tfrac{25}{36}\sin\!\big(\tfrac{\pi}{6}-\tfrac{5u}{6}\big).$$

For $0 \le u \le \pi/7$ the first argument lies in $[\frac{\pi}{6}, \frac{\pi}{3}]$, so its sine is $\ge \frac{1}{2}$; the second lies in $[\frac{\pi}{21}, \frac{\pi}{6}]$, so its sine is positive. Hence $S'' \le -\frac{49}{72} < 0$ and S is strictly concave on $[0, \pi/7]$.

(i) $S(0) = 2\sin\frac{\pi}{6} = 1$. At $u = \pi/8$ the arguments are $\frac{\pi}{6}+\frac{7\pi}{48} = \frac{5\pi}{16}$ and $\frac{\pi}{6}-\frac{5\pi}{48} = \frac{\pi}{16}$, so $S(\pi/8) = \sin\frac{5\pi}{16} + \sin\frac{\pi}{16} = 0.8315\ldots + 0.1951\ldots = 1.0266\ldots > 1$. A strictly concave function lies strictly above its chord, so $S > 1$ on $(0, \pi/8]$.

(ii) Put $T(u) = S(u) - \cos\frac{u}{2}$. Then $T'' = S'' + \frac{1}{4}\cos\frac{u}{2} \le -\frac{49}{72} + \frac{1}{4} = -0.4306\ldots < 0$, so T is strictly concave on $[0, \pi/7]$. $T(0) = 1 - 1 = 0$. At $u = \pi/7$ the arguments are $\frac{\pi}{6}+\frac{7\pi}{42} = \frac{\pi}{3}$ and $\frac{\pi}{6}-\frac{5\pi}{42} = \frac{\pi}{21}$, so $T(\pi/7) = \sin\frac{\pi}{3} + \sin\frac{\pi}{21} - \cos\frac{\pi}{14} = 0.8660\ldots + 0.1490\ldots - 0.9749\ldots = 0.0401\ldots > 0$. Hence $T > 0$ on $(0, \pi/7]$. □

Theorem 3.2 (Spacing)**.** *Let $n \ge 7$ and let e, f be edges with cyclic distance $d = d(e,f)$. Then*

(a) if $d < n/6$, the interiors of N_e and N_f overlap;

(b) if $d > n/6$, they do not overlap, and in fact $\operatorname{dist}(N_e, N_f) > 0$;

(c) if $d = n/6$ (so $6 \mid n$), they do not overlap but touch.

Consequently d is admissible if and only if $d \ge \lceil n/6 \rceil$, i.e. $m(n) = \lceil n/6 \rceil$.

Proof. Write $c = c_{ef}$. By Corollary 2.3 we must decide whether $c \in \operatorname{int} D$, using (1) and the bounds $\{|x| \le r_D\} \subseteq D \subseteq \{|x| \le R_D\}$ of Lemma 2.4.

(a) Let $d < n/6$. Then $0 < ud < \pi/6 < \pi/2$ and sin is increasing there, so $|c| = 4a\sin(ud) < 4a\sin\frac{\pi}{6} = 2a$. Now $r_D = 2a$ for even n and $r_D = 1 + a \ge 2a$ for odd n (because $a \le 1$). So $|c| < r_D$, hence c lies in the open disc of radius r_D, which is contained in $\operatorname{int} D$.

(b) Let d_0 be the smallest integer with $d_0 > n/6$, and write $d_0 = \frac{n}{6} + \eta_0$. If $n = 6q + r$ with $1 \le r \le 5$, then $d_0 = q + 1$ and $\eta_0 = (q+1) - \frac{6q+r}{6} = \frac{6-r}{6} \in [\frac{1}{6}, \frac{5}{6}]$. If $6 \mid n$, then $d_0 = \frac{n}{6} + 1$ and $\eta_0 = 1$. In all cases $\frac{1}{6} \le \eta_0 \le 1$. For every d with $d_0 \le d \le n/2$ we have $\frac{\pi d}{n} \le \frac{\pi}{2}$, and since sin is increasing on $[0, \frac{\pi}{2}]$,

$$|c| = 4a\sin\tfrac{\pi d}{n} \;\ge\; 4a\sin\big(\tfrac{\pi}{6} + \eta_0 u\big) \;\ge\; 4a\sin A_0, \qquad A_0 := \tfrac{\pi}{6} + \tfrac{u}{6},$$

where we used $\frac{\pi d_0}{n} = \frac{\pi}{6} + \eta_0 u$ and $\eta_0 \ge \frac{1}{6}$ (the argument stays below $\frac{\pi}{6} + u \le \frac{\pi}{6} + \frac{\pi}{7} < \frac{\pi}{2}$). The product-to-sum identity $2\cos u \sin A = \sin(A+u) + \sin(A-u)$ (add $\sin(A \pm u) = \sin A\cos u \pm \cos A \sin u$) with $A = A_0$ gives, since $A_0 + u = \frac{\pi}{6} + \frac{7u}{6}$ and $A_0 - u = \frac{\pi}{6} - \frac{5u}{6}$,

$$4a\sin A_0 = 2 \cdot 2\cos u \sin A_0 = 2\,S(u), \qquad S(u) = \sin\!\big(\tfrac{\pi}{6}+\tfrac{7u}{6}\big) + \sin\!\big(\tfrac{\pi}{6}-\tfrac{5u}{6}\big).$$

- n even (so $n \ge 8$ and $u \le \pi/8$): $|c| \ge 2S(u) > 2 = R_D$ by Lemma 3.1(i).
- n odd (so $u \le \pi/7$): $|c| \ge 2S(u) > 2\cos\frac{u}{2} = R_D$ by Lemma 3.1(ii).

In both cases $|c| > R_D$, so $c \notin D$ (because $D \subseteq \{|x| \le R_D\}$). Thus the interiors are disjoint and $\operatorname{dist}(N_e, N_f) = \operatorname{dist}(c, D) > 0$ since D is closed.

(c) Let $n = 6q$ and $d = q$; by rotation take $e = 0$, $f = q$. For angles α, β the difference formulas $\cos\beta - \cos\alpha = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\beta-\alpha}{2}$ and $\sin\beta - \sin\alpha = 2\cos\frac{\alpha+\beta}{2}\sin\frac{\beta-\alpha}{2}$ give

$$(\cos\beta - \cos\alpha,\ \sin\beta - \sin\alpha) = 2\sin\tfrac{\beta-\alpha}{2}\,(\cos\gamma,\ \sin\gamma), \qquad \gamma = \tfrac{\alpha+\beta}{2} + \tfrac{\pi}{2}.$$

Take $\alpha = \psi_0 = u$ and $\beta = \psi_q = (2q+1)u$. Then $\frac{\beta-\alpha}{2} = qu = \frac{\pi}{6}$, so $2\sin\frac{\beta-\alpha}{2} = 1$; and since $\frac{\pi}{2} = \frac{n}{2}u = 3qu$, $\gamma = (q+1)u + 3qu = (4q+1)u = \psi_{2q}$. Hence $u_q - u_0 = (\cos\psi_{2q}, \sin\psi_{2q}) = u_{2q}$ and

$$c = 2a\,u_{2q}.$$

Because n is even, $-P = P$ and $D = 2P$, whose support function in the direction ψ_{2q} is $2h_P(\psi_{2q}) = 2a$ (the direction ψ_{2q} is an edge normal, midway between vertex directions, so $\delta = u$ and $h_P = \cos u = a$). So $\langle c, u_{2q}\rangle = 2a = h_D(\psi_{2q})$: the point c lies on a supporting line of D, hence

$c \notin \operatorname{int} D$, and it is the midpoint of the corresponding edge of $2P$, hence $c \in D$. The interiors are disjoint and the neighbours touch.

Finally, (a)–(c) say that d is admissible iff $d \geq n/6$, i.e. (for integers) iff $d \geq \lceil n/6 \rceil$. □

Proposition 3.3 (Contact when $6 \mid n$: a shared edge)**.** *Let $n = 6q$. The neighbours N_0 and N_q intersect in a full edge of length ℓ (the "perfect ring" contact of the GeoGebra drawing of the 36-gon).*

Proof. n is even, so $-P = P$ and $N_e = O_e + P$ with $O_e = 2a\,u_e$. By part (c) above, $N_0 \cap N_q = O_0 + \big(P \cap (P + c)\big)$ with $c = 2a\,w$, $w := u_{2q}$. $P \subseteq \{\langle x, w\rangle \leq a\}$ and, by central symmetry, $P \subseteq \{\langle x, w\rangle \geq -a\}$, so $P + c \subseteq \{\langle x, w\rangle \geq a\}$. The intersection therefore lies on the line $\langle x, w\rangle = a$. Let $F = P \cap \{\langle x, w\rangle = a\} = \{aw + t\omega : \ |t| \leq \ell/2\}$ be the edge of P with normal w ($\omega \perp w$ a unit vector). By symmetry $P \cap \{\langle x, w\rangle = -a\} = -F$, so $(P + c) \cap \{\langle x, w\rangle = a\} = c - F = \{2aw - aw - t\omega\} = \{aw + t'\omega : \ |t'| \leq \ell/2\} = F$. Thus $P \cap (P + c) = F$, an edge of length ℓ. □

Proposition 3.4 (Small polygons)**.** *For $3 \leq n \leq 6$ no two neighbours overlap, so $m(n) = 1$.*

Proof. Compare $|c_d| = 4a \sin \frac{\pi d}{n}$ with R_D (Lemma 2.4); $|c| \geq R_D$ implies $c \notin \operatorname{int} D$.

- $n = 3$ ($R_D = 2\cos\frac{\pi}{6} = \sqrt{3}$): $d = 1$: $|c| = 4 \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \sqrt{3} = R_D$.
- $n = 4$ ($R_D = 2$): $d = 1$: $|c| = 4 \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} = 2 = R_D$; $d = 2$: $|c| = 4 \cdot \frac{\sqrt{2}}{2} = 2\sqrt{2} > R_D$.
- $n = 5$ ($R_D = 2\cos 18^\circ = 1.9021\ldots$): $d = 1$: $|c| = 4\cos 36^\circ \sin 36^\circ = 2\sin 72^\circ = 1.9021\ldots = R_D$; $d = 2$: $|c| = 4\cos 36^\circ \sin 72^\circ = 3.0777\ldots > R_D$.
- $n = 6$ ($R_D = 2$): $d = 1$: the touching case (c) of Theorem 3.2, whose proof uses only $n = 6q$, so it applies with $q = 1$; $d = 2$: $|c| = 4 \cdot \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = 3 > R_D$; $d = 3$: $|c| = 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3} > R_D$.

In every case $c \notin \operatorname{int} D$. □

Corollary 3.5 (Characterisation of placements)**.** *For every $n \geq 3$ and $S \subseteq \mathbb{Z}_n$: S is a placement $\iff$ any two distinct elements of S have cyclic distance $\geq m(n)$ $\iff$ (when $|S| \geq 2$) all gaps of S are $\geq m(n)$.*

Proof. The first equivalence is Theorem 3.2 together with Proposition 3.4 (for $n \leq 6$, $m = 1$ and every distance is admissible). Suppose all pairwise cyclic distances are $\geq m$. A gap g_i is the length of one of the two arcs between two consecutive elements, and the cyclic distance of these elements is $\min(g_i, n - g_i) \geq m$, so $g_i \geq m$. Conversely, suppose all gaps are $\geq m$. Two distinct elements of S are separated, along each of the two arcs joining them, by at least one gap, so both arcs have length $\geq m$ and the cyclic distance is $\geq m$. □

4 Conjoinability, slack and vertex skipping: formulas and values

Theorem 4.1 (Formulas)**.** *For every $n \geq 3$,*

$$m(n) = \left\lceil \frac{n}{6} \right\rceil, \qquad k(n) = \left\lfloor \frac{n}{m(n)} \right\rfloor,$$

$$s(n) = \max\big(0,\ m(n) - 2\big), \quad \sigma(n) = n - k(n)\,m(n) = n \bmod m(n) \in \{0, \ldots, m(n) - 1\}.$$

Proof. For $n \geq 7$, $m(n) = \lceil n/6 \rceil$ is Theorem 3.2. For $3 \leq n \leq 6$, Proposition 3.4 gives $m(n) = 1 = \lceil n/6 \rceil$. By Corollary 3.5, a placement of size $t \geq 2$ is a choice of t gaps $g_1, \ldots, g_t \geq m$ with $g_1 + \cdots + g_t = n$ (the edges are recovered as partial sums), and such gaps exist if and only if $tm \leq n$. A placement of size 1 always exists. Hence the largest size is $\lfloor n/m \rfloor$ (note $n/m \geq 3 > 1$). The formula for s is Definition 1.5. Finally $\sigma = n - \lfloor n/m \rfloor m$ is the remainder of n on division by m, so $0 \leq \sigma \leq m - 1$. □

Theorem 4.2 (Values of k, σ and s)**.** *Write $n = 6q + r$ with $q = \lfloor n/6 \rfloor$ and $r = n \bmod 6 \in \{0, 1, \ldots, 5\}$.*

(a) $3 \le n \le 6$. $m = 1$, $k(n) = n$, $\sigma = 0$, $s = 0$.

(b) $6 \mid n$ ($r = 0$, $q \ge 1$). $m = q$, $k(n) = 6$, $\sigma = 0$, $s = \max(0, q-2)$.

(c) $n \ge 7$, $6 \nmid n$ ($1 \le r \le 5$, $q \ge 1$). $m = q+1$, $s = q-1$, *and*

$$k(n) = 6 - \left\lceil \frac{6-r}{q+1} \right\rceil.$$

In particular $k(n) = 5$ *and* $\sigma(n) = q + r - 5 = \lfloor n/6 \rfloor + (n \bmod 6) - 5$ *for every such* n except *the six values*

n	7	8	9	13	14	19
$k(n)$	3	4	4	4	4	4
$\sigma(n)$	1	0	1	1	2	3

Table 2 lists the values for $3 \le n \le 38$.

n	m	k	σ	s	n	m	k	σ	s	n	m	k	σ	s
3	1	3	0	0	15	3	5	0	1	27	5	5	2	3
4	1	4	0	0	16	3	5	1	1	28	5	5	3	3
5	1	5	0	0	17	3	5	2	1	29	5	5	4	3
6	**1**	**6**	**0**	**0**	**18**	**3**	**6**	**0**	**1**	**30**	**5**	**6**	**0**	**3**
7	2	3	1	0	19	4	4	3	2	31	6	5	1	4
8	2	4	0	0	20	4	5	0	2	32	6	5	2	4
9	2	4	1	0	21	4	5	1	2	33	6	5	3	4
10	2	5	0	0	22	4	5	2	2	34	6	5	4	4
11	2	5	1	0	23	4	5	3	2	35	6	5	5	4
12	**2**	**6**	**0**	**0**	**24**	**4**	**6**	**0**	**2**	**36**	**6**	**6**	**0**	**4**
13	3	4	1	1	25	5	5	0	3	37	7	5	2	5
14	3	4	2	1	26	5	5	1	3	38	7	5	3	5

Table 2: Minimal spacing $m = \lceil n/6 \rceil$, conjoinability $k = \lfloor n/m \rfloor$, slack $\sigma = n - km$ and vertex skipping $s = \max(0, m-2)$ for $3 \le n \le 38$ (multiples of 6 in bold).

Proof. (a) was shown in Theorem 4.1: $m = 1$ gives $k = \lfloor n/1 \rfloor = n$ and $\sigma = n - n = 0$; $s = \max(0, -1) = 0$.

(b) If $n = 6q$ then $m = \lceil q \rceil = q$ and $k = \lfloor 6q/q \rfloor = 6$, $\sigma = 6q - 6q = 0$, $s = \max(0, q-2)$.

(c) If $1 \le r \le 5$ then $6q < n < 6(q+1)$, so $m = q+1$ and $s = m - 2 = q - 1$. Since $6(q+1) = 6q + 6$ we can write $n = 6(q+1) - (6-r)$ and

$$\frac{n}{m} = \frac{6(q+1) - (6-r)}{q+1} = 6 - X, \qquad X := \frac{6-r}{q+1} > 0.$$

For the integer 6 we have $\lfloor 6 - X \rfloor = 6 - \lceil X \rceil$, hence $k = 6 - \lceil X \rceil$. Now $X \le 1 \iff q + 1 \ge 6 - r$, and then $k = 5$.

- $q \ge 4$: $q + 1 \ge 5 \ge 6 - r$ for all $r \ge 1$, so $k = 5$.
- $q = 3$ ($n = 19, \ldots, 23$): $X = \frac{6-r}{4}$. $r = 1$: $X = \frac{5}{4} \in (1, 2]$, $k = 4$ ($n = 19$). $r \ge 2$: $X \le 1$, $k = 5$.
- $q = 2$ ($n = 13, \ldots, 17$): $X = \frac{6-r}{3}$. $r = 1, 2$: $X = \frac{5}{3}, \frac{4}{3} \in (1, 2]$, $k = 4$ ($n = 13, 14$). $r \ge 3$: $X \le 1$, $k = 5$.
- $q = 1$ ($n = 7, \ldots, 11$): $X = \frac{6-r}{2}$. $r = 1$: $X = \frac{5}{2} \in (2, 3]$, $k = 3$ ($n = 7$). $r = 2, 3$: $X = 2, \frac{3}{2} \in (1, 2]$, $k = 4$ ($n = 8, 9$). $r = 4, 5$: $X = 1, \frac{1}{2}$, $k = 5$ ($n = 10, 11$).

For $k = 5$: $\sigma = n - 5m = (6q + r) - 5(q+1) = q + r - 5$. For the exceptions $\sigma = n - km$ directly: $n = 7$: $m = 2$, $7 - 3 \cdot 2 = 1$; $n = 8$: $m = 2$, $8 - 4 \cdot 2 = 0$; $n = 9$: $9 - 8 = 1$; $n = 13$: $m = 3$, $13 - 4 \cdot 3 = 1$; $n = 14$: $14 - 12 = 2$; $n = 19$: $m = 4$, $19 - 4 \cdot 4 = 3$. □

Proposition 4.3 (Hexagonal resonance of the conjoinability)**.** *For every $n \geq 3$, $k(n) \leq 6$, and for $n \geq 6$: $k(n) = 6 \iff 6 \mid n$. Moreover $k(n) = 5$ for every $n \geq 20$ with $6 \nmid n$.*

Proof. From Theorem 4.2: $k = n \leq 5$ for $n = 3, 4, 5$; $k = 6$ for $6 \mid n$; and for $6 \nmid n$, $n \geq 7$, $k = 6 - \lceil X \rceil \leq 5$ because $X > 0$. The last claim: $n \geq 20$, $6 \nmid n$ means $q \geq 3$, and by Theorem 4.2(c) the only values with $k < 5$ are $n \leq 19$. □

Corollary 4.4 (Slack zero and slack one)**.** *For $n \geq 7$:*
(a) $\sigma(n) = 0 \iff 6 \mid n$ or $n \in \{8, 10, 15, 20, 25\}$;
(b) $\sigma(n) = 1 \iff n \in \{7, 9, 11, 13, 16, 21, 26, 31\}$;
(c) $\sigma(n) \geq 2$ for every $n \geq 32$ with $6 \nmid n$, and $\sigma(n) \to \infty$ along such n.

Proof. By Theorem 4.1, $\sigma \equiv n \pmod{m}$ with $0 \leq \sigma < m$. For $6 \nmid n$ write $n = 6m - (6 - r)$ with $m = q + 1 \geq 2$ and $1 \leq 6 - r \leq 5$ (every such pair (m, r) gives $\lceil n/6 \rceil = m$).

(a) $\sigma = 0 \iff m \mid n \iff m \mid (6 - r)$. With $m \geq 2$ and $6 - r \in \{1, \ldots, 5\}$: $6 - r = 5$ gives $m = 5$ ($n = 25$); $6 - r = 4$ gives $m \in \{2, 4\}$ ($n = 8, 20$); $6 - r = 3$ gives $m = 3$ ($n = 15$); $6 - r = 2$ gives $m = 2$ ($n = 10$); $6 - r = 1$ has no divisor ≥ 2. For $6 \mid n$, $\sigma = 0$ by Theorem 4.2(b).

(b) $\sigma = 1$ with $m \geq 2$ means $n \equiv 1 \pmod{m}$, i.e. $m \mid n - 1 = 6m - (7 - r)$, i.e. $m \mid 7 - r$ with $7 - r \in \{2, \ldots, 6\}$ (for $6 \mid n$ the slack is 0). $7 - r = 6$ ($r = 1$): $m \in \{2, 3, 6\}$, $n = 6m - 5 = 7, 13, 31$. $7 - r = 5$ ($r = 2$): $m = 5$, $n = 26$. $7 - r = 4$ ($r = 3$): $m \in \{2, 4\}$, $n = 6m - 3 = 9, 21$. $7 - r = 3$ ($r = 4$): $m = 3$, $n = 16$. $7 - r = 2$ ($r = 5$): $m = 2$, $n = 11$.

(c) For $6 \nmid n$, $n \geq 32$ we have $q \geq 5$ and, by Proposition 4.3, $k = 5$, so $\sigma = q + r - 5$. If $q \geq 6$ then $q + r - 5 \geq 7 - 5 = 2$; if $q = 5$ then $n = 31 + (r - 1)$ and $r \geq 2$ (since $n \geq 32$) gives $q + r - 5 \geq 2$. Also $q + r - 5 \geq q - 4 \to \infty$. □

5 Counting placements

Throughout, $m = m(n)$, $k = k(n)$ and $\sigma = \sigma(n)$.

Theorem 5.1 (Number of placements)**.** *For $1 \leq t \leq n/m$ the number of placements of size t is*

$$N(n, t) = \frac{n}{t} \binom{n - t(m-1) - 1}{t - 1},$$

and it is 0 for $t > n/m$. (We also put $N(n, 0) = 1$.)

Proof. Let $\mathcal{P}_t$ be the set of placements of size t and Comp the set of compositions $(g_1, \ldots, g_t)$ of n with every $g_i \geq m$. Send a pair (S, e) with $S \in \mathcal{P}_t$ and $e \in S$ to $(e, (g_1, \ldots, g_t))$, where g_i are the gaps met when walking from e once around $\mathbb{Z}_n$ in the positive direction. By Corollary 3.5 this is a bijection onto $\mathbb{Z}_n \times \mathrm{Comp}$ (the inverse takes partial sums). Each S occurs in exactly t pairs, so $t\,|\mathcal{P}_t| = n\,|\mathrm{Comp}|$. Subtracting $m - 1$ from every part turns Comp into the compositions of $n - t(m - 1)$ into t positive parts, of which there are $\binom{n-t(m-1)-1}{t-1}$ (stars and bars). □

Example: all counts for $n = 34$. Here $m = 6$, $k = 5$, $\sigma = 4$, and $n - t(m - 1) - 1 = 34 - 5t - 1 = 33 - 5t$:

$$N(34, 1) = \tfrac{34}{1}\tbinom{28}{0} = 34, \qquad N(34, 2) = \tfrac{34}{2}\tbinom{23}{1} = 17 \cdot 23 = 391,$$
$$N(34, 3) = \tfrac{34}{3}\tbinom{18}{2} = \tfrac{34}{3} \cdot 153 = 1734, \quad N(34, 4) = \tfrac{34}{4}\tbinom{13}{3} = \tfrac{34}{4} \cdot 286 = 2431,$$
$$N(34, 5) = \tfrac{34}{5}\tbinom{8}{4} = \tfrac{34 \cdot 70}{5} = 476.$$

For $n = 6q$: $N(6q, 6) = \frac{6q}{6}\binom{6q - 6(q-1) - 1}{5} = q\binom{5}{5} = q$: the extremal placements are exactly the q rotations of the perfect ring.

Proposition 5.2 (Placements up to symmetry)**.** *Two placements of size t differ by a rotation of P if and only if their cyclic gap sequences differ by a cyclic shift, and by a rotation or a reflection if and only if they differ by a cyclic shift and/or a reversal. Hence the number $R(n,t)$ of placements of size t up to rotation, and the number $B(n,t)$ up to rotation and reflection, are the numbers of necklaces and of bracelets of compositions of n into t parts $\geq m$. By Burnside's lemma,*

$$R(n,t) = \frac{1}{t}\sum_{j=0}^{t-1} F_j, \qquad B(n,t) = \frac{1}{2t}\Big(\sum_{j=0}^{t-1} F_j + G\Big),$$

where, for $h = \gcd(j,t)$ and $w = t/h$ (with $\gcd(0,t) = t$),

$$F_j = \begin{cases} \binom{n/w - h(m-1) - 1}{h-1} & \text{if } w \mid n \text{ and } n/w \geq hm, \\ 0 & \text{otherwise,} \end{cases}$$

$$G = \begin{cases} t\, W(n; 1, \frac{t-1}{2}) & t \text{ odd}, \\ \frac{t}{2}\big(W(n; 2, \frac{t-2}{2}) + W(n; 0, \frac{t}{2})\big) & t \text{ even}, \end{cases}$$

and $W(n;a,b)$ is the number of pairs $\big((x_1,\dots,x_a),(y_1,\dots,y_b)\big)$ of integers $\geq m$ with $x_1 + \cdots + x_a + 2(y_1 + \cdots + y_b) = n$.

Proof. Translating S along $\mathbb{Z}_n$ keeps its cyclic gap sequence, and conversely two placements with the same cyclic gap sequence are translates of each other; a reflection $e \mapsto -e$ reverses the gap sequence. This gives the first two sentences. The cyclic group C_t (shifts) and the dihedral group D_t (shifts and reversals) act on Comp, and the counts are the numbers of orbits; apply Burnside's lemma $|\text{orbits}| = \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma} |\text{Fix}(\gamma)|$. A composition fixed by the shift by j is periodic with period $h = \gcd(j,t)$, so it is determined by its first h parts, which sum to n/w (this requires $w \mid n$) and are each $\geq m$; there are $\binom{n/w - h(m-1) - 1}{h-1}$ of them. This is F_j. A reflection of the index set acts as $i \mapsto b - i$ on $\mathbb{Z}_t$. For t odd every such reflection has exactly one fixed index and $\frac{t-1}{2}$ pairs $\{i, b-i\}$; for t even, $\frac{t}{2}$ reflections have two fixed indices and $\frac{t-2}{2}$ pairs, and $\frac{t}{2}$ reflections have no fixed index and $\frac{t}{2}$ pairs. A fixed composition has equal parts y on each pair and arbitrary parts x on the fixed indices, so the sum is $\sum x + 2\sum y = n$. Counting these gives W. □

Worked example: $n = 34$**,** $t = 5$ **(**$m = 6$**).** *Rotations.* For $j = 0$: $h = 5$, $w = 1$, so $F_0 = \binom{34 - 5\cdot 5 - 1}{4} = \binom{8}{4} = 70$. For $j = 1,2,3,4$: $h = 1$, $w = 5$, and $5 \nmid 34$, so $F_j = 0$. Hence $R(34,5) = \frac{70}{5} = 14$. *Reflections* ($t = 5$ odd): $W(34;1,2)$ counts $(x; y_1, y_2)$ with $x, y_1, y_2 \geq 6$ and $x + 2(y_1 + y_2) = 34$. Put $Y = y_1 + y_2 \geq 12$; then $x = 34 - 2Y \geq 6$ forces $Y \leq 14$. $Y = 12$: $(y_1, y_2) = (6,6)$, 1 pair, $x = 10$. $Y = 13$: $(6,7), (7,6)$, 2 pairs, $x = 8$. $Y = 14$: $(6,8), (7,7), (8,6)$, 3 pairs, $x = 6$. So $W = 1 + 2 + 3 = 6$ and $G = 5 \cdot 6 = 30$. Hence $B(34,5) = \frac{70+30}{2\cdot 5} = 10$.

Theorem 5.3 (Rigidity of the extremal placement)**.** *Let $n \geq 7$. The number of extremal placements (size k) up to rotation equals 1 if and only if $\sigma \leq 1$. In particular, if $6 \mid n$ the extremal placement is unique up to rotation (the perfect ring).*

Proof. Write the gaps as $g_i = m + \epsilon_i$ with $\epsilon_i \geq 0$ and $\sum_{i=1}^{k} \epsilon_i = n - km = \sigma$. Up to rotation, extremal placements correspond to necklaces of such vectors $(\epsilon_1, \dots, \epsilon_k)$. If $\sigma = 0$ the vector is zero: one necklace. If $\sigma = 1$ the vector has a single entry 1 and zeros elsewhere, and all its cyclic shifts are equivalent: one necklace. If $\sigma \geq 2$ (recall $k \geq 3$ for $n \geq 7$), the vectors $(\sigma, 0, \dots, 0)$ and $(\sigma - 1, 1, 0, \dots, 0)$ have different multisets of entries (the largest entry is σ in the first and $\sigma - 1$ in the second, since $\sigma - 1 \geq 1$), so they are not cyclic shifts of each other: there are at least two necklaces. If $6 \mid n$ then $\sigma = 0$ by Theorem 4.2(b). □

Corollary 5.4 (The rigid values of n)**.** *For $n \geq 7$ the extremal placement is unique up to rotation if and only if $6 \mid n$ or $n \in \{7, 8, 9, 10, 11, 13, 15, 16, 20, 21, 25, 26, 31\}$.*

Proof. Theorem 5.3 gives uniqueness iff $\sigma \le 1$, and Corollary 4.4 lists these n: slack 0 for $6 \mid n$ and $n \in \{8, 10, 15, 20, 25\}$, slack 1 for $n \in \{7, 9, 11, 13, 16, 21, 26, 31\}$. □

Theorem 5.5 (Tight chains with BA fixed)**.** *Fix the neighbour on edge* 0 *("BA"). Call a placement of size k containing the edge* 0 *a* tight chain *if all its gaps equal m except one, the* pocket*, of size $m+\sigma$. Identify a placement with its mirror image in the axis through BA. The number of tight chains is* 1 *if $\sigma = 0$ and $\lceil k/2 \rceil$ if $\sigma > 0$.*

Proof. If $\sigma = 0$ all gaps equal m and there is one chain. Let $\sigma > 0$. Counting the gaps from BA, the pocket can be the j-th gap, $j = 0, 1, \dots, k-1$, and different choices give different placements containing the edge 0. The reflection $e \mapsto -e$ of $\mathbb{Z}_n$ fixes the edge 0 and reverses the gap sequence, so it maps position j to $k-1-j$. Its orbits are the pairs $\{j, k-1-j\}$ with $j \ne k-1-j$ and, if k is odd, the single fixed position $j = \frac{k-1}{2}$. The number of orbits is $\lceil k/2 \rceil$. □

Example ($n = 34$). Here $k = 5$, so $\lceil 5/2 \rceil = 3$. Writing $a{+}b$ for the numbers of tight neighbours met on the two sides of BA before the pocket, the three chains are 4+0 (edges $0, 6, 12, 18, 24$), 3+1 $(0, 6, 12, 18, 28)$ and 2+2 $(0, 6, 12, 22, 28)$, each with a pocket gap of $m + \sigma = 6 + 4 = 10$ edges, i.e. 8 untouched central vertices. The first two are the GeoGebra files `34_placement_1` and `34_placement_2`; all three are rotations of one arrangement (Figure 1).

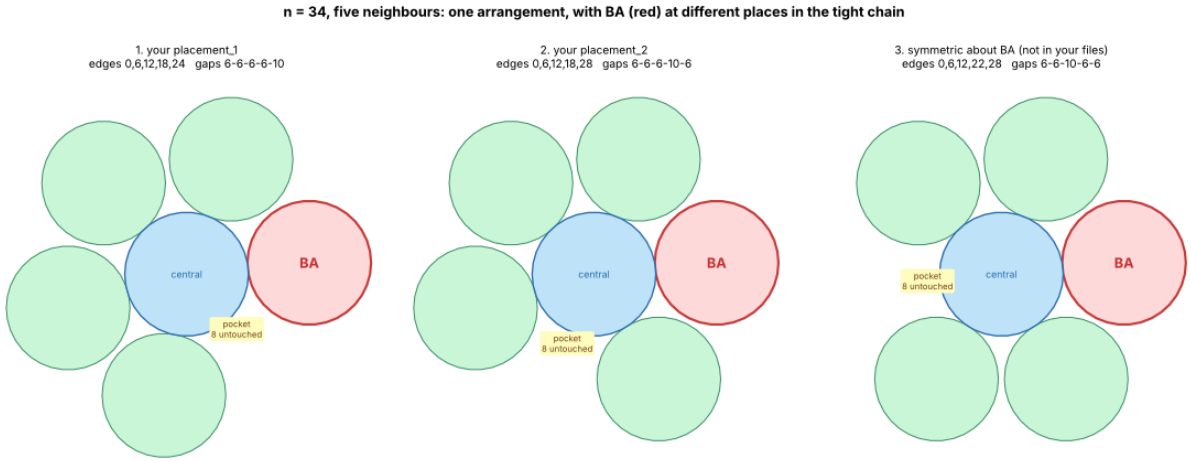


Figure 1: The three tight chains with BA fixed for $n = 34$ (BA in red).

6 A generating function for placements (hypothesis H3)

For integers $n \ge m \ge 1$ write $N_m(n, t) = \frac{n}{t}\binom{n-t(m-1)-1}{t-1}$ for $t \ge 1$ and $N_m(n, 0) = 1$, and define the *placement polynomial*

$$F_{n,m}(x) = \sum_{t \ge 0} N_m(n, t)\, x^t,$$

the generating polynomial for subsets of $\mathbb{Z}_n$ whose elements are pairwise at cyclic distance $\ge m$ (Theorem 5.1). For the n-gon take $m = m(n)$ and write $P_n(x) := F_{n,m(n)}(x)$; its coefficients are the numbers of placements by size and $\deg P_n = k(n)$. For example $P_{34}(x) = 1 + 34x + 391x^2 + 1734x^3 + 2431x^4 + 476x^5$ (the numbers computed after Theorem 5.1).

Theorem 6.1 (Placement polynomial as a power sum)**.** *Let $m \ge 1$ and let $\lambda_1(x), \dots, \lambda_m(x)$ be the roots of $\lambda^m = \lambda^{m-1} + x$, and $p_n(x) = \sum_{i=1}^{m} \lambda_i(x)^n$. Then*

(a) $$\sum_{n \ge 1} p_n(x)\, z^n = \frac{z\,(1 + m\,x\,z^{m-1})}{1 - z - x\,z^m} = -z\,\frac{\partial}{\partial z} \log\left(1 - z - xz^m\right);$$

(b) $F_{n,m}(x) = p_n(x)$ *for all* $n \geq m$*;*
(c) $F_{n,m} = F_{n-1,m} + x\,F_{n-m,m}$ *for all* $n \geq 2m$*.*

Proof. (a) Let $\chi(\lambda) = \lambda^m - \lambda^{m-1} - x = \prod_{i=1}^{m}(\lambda - \lambda_i)$. Then

$$z^m\chi(1/z) = z^m\big(z^{-m} - z^{-(m-1)} - x\big) = 1 - z - xz^m,$$
$$z^m\chi(1/z) = \prod_i z\,(1/z - \lambda_i) = \prod_i (1 - \lambda_i z),$$

so $1 - z - xz^m = \prod_i(1 - \lambda_i z)$. Taking the logarithmic derivative, and expanding each geometric series,

$$-z\frac{\partial}{\partial z}\log\prod_i(1-\lambda_i z) = \sum_i \frac{\lambda_i z}{1-\lambda_i z} = \sum_i\sum_{n\geq 1}\lambda_i^n z^n = \sum_{n\geq 1} p_n z^n.$$

On the other hand $\frac{\partial}{\partial z}(1-z-xz^m) = -(1+mxz^{m-1})$, so the same expression equals $\frac{z(1+mxz^{m-1})}{1-z-xz^m}$.

(b) *Step 1.* Write $1 - z - xz^m = (1-z)\big(1 - \frac{xz^m}{1-z}\big)$ and expand the geometric series in x:

$$\frac{1}{1-z-xz^m} = \sum_{t\geq 0}\frac{x^t z^{mt}}{(1-z)^{t+1}}.$$

Step 2. Multiply by $z + mxz^m$ and re-index the second sum $(t+1 \to t)$:

$$\frac{z(1+mxz^{m-1})}{1-z-xz^m} = \sum_{t\geq 0}\frac{x^t z^{mt+1}}{(1-z)^{t+1}} + m\sum_{t\geq 1}\frac{x^t z^{mt}}{(1-z)^t}.$$

Step 3. Use $[z^j](1-z)^{-\rho} = \binom{j+\rho-1}{\rho-1}$ for $j \geq 0$ (and 0 for $j < 0$), and put $M := n - t(m-1) - 1$. Then

$$[z^n]\frac{z^{mt+1}}{(1-z)^{t+1}} = \binom{n-mt-1+t}{t} = \binom{M}{t},$$
$$m\,[z^n]\frac{z^{mt}}{(1-z)^t} = m\binom{n-mt+t-1}{t-1} = m\binom{M}{t-1},$$

where a binomial coefficient $\binom{M}{j}$ is read as 0 when $M < j$ (this covers the cases $n < mt+1$ resp. $n < mt$). Hence for $t \geq 1$ the coefficient of $x^t z^n$ is $\binom{M}{t} + m\binom{M}{t-1}$. Since $\binom{M}{t} = \binom{M}{t-1}\frac{M-t+1}{t}$,

$$\binom{M}{t} + m\binom{M}{t-1} = \binom{M}{t-1}\frac{M-t+1+mt}{t} = \binom{M}{t-1}\frac{n}{t} = N_m(n,t),$$

because $M - t + 1 + mt = \big(n - tm + t - 1\big) - t + 1 + mt = n$. For $t = 0$ only the first sum contributes and $[z^n]\frac{z}{1-z} = 1 = N_m(n,0)$ for $n \geq 1$. So $[x^t z^n]\sum_n p_n z^n = N_m(n,t)$ for all t and $n \geq m$, which is (b).

(c) Multiply $\lambda_i^m = \lambda_i^{m-1} + x$ by λ_i^{n-m} to get $\lambda_i^n = \lambda_i^{n-1} + x\lambda_i^{n-m}$, and sum over i: $p_n = p_{n-1} + x\,p_{n-m}$. For $n \geq 2m$ both $n-1$ and $n-m$ are $\geq m$, so (b) turns this into the claim. □

Example $(m = 2)$**.** The roots of $\lambda^2 = \lambda + x$ satisfy $\lambda_1 + \lambda_2 = 1$, $\lambda_1\lambda_2 = -x$, so $p_1 = 1$, $p_2 = 1 + 2x$, and by (c) $p_n = p_{n-1} + xp_{n-2}$: $p_3 = 1 + 3x$, $p_4 = 1 + 4x + 2x^2$, $p_5 = 1 + 5x + 5x^2$. This agrees with $N_2(5,1) = 5$, $N_2(5,2) = \frac{5}{2}\binom{2}{1} = 5$ (subsets of the 5-cycle with no two adjacent elements).

Corollary 6.2 ("Re-entry into $k = 6$")**.** *The* n*-gons with minimal spacing* $m(n) = m \geq 2$ *are exactly* $n = 6m - 5, 6m - 4, \ldots, 6m$*. On this window* $P_n = F_{n,m}$ *and* $\deg P_n = \lfloor n/m \rfloor$*, which equals* 6 *only at the right end* $n = 6m$*, where* $[x^6]P_{6m} = m$*. Thus* $[x^6]P_n \neq 0 \iff 6 \mid n$*, and the full sequence* (P_n) *is obtained by concatenating the six-term windows of the rational functions* $\dfrac{z(1+mxz^{m-1})}{1-z-xz^m}$*,* $m = 2, 3, \ldots$

Proof. $\lceil n/6\rceil = m \iff 6(m-1) < n \le 6m$. On the window, $\lfloor n/m\rfloor = 6$ iff $n \ge 6m$, i.e. $n = 6m$. The coefficient is $N_m(6m,6) = \frac{6m}{6}\binom{6m-6(m-1)-1}{5} = m\binom{5}{5} = m$. The remaining claims follow from Theorems 4.1, 5.1 and 6.1. □

Remark 6.3. This is the exact content that the sentence "a generating function $P(x,y)$ models the periodic re-entry into $k = 6$ states" can have. The one-variable generating function of the conjoinability itself is trivial: $k(n) = 5 + [6 \mid n]$ for $n \ge 20$, so $\sum_n k(n)z^n$ is a polynomial plus $\frac{5z^{20}}{1-z} + \frac{z^{24}}{1-z^6}$.

7 Tension and the discrete-to-continuous profile (hypotheses H4, H5)

Recall Definition 1.6: the tension $\tau(n) = \operatorname{dist}(N_e, N_f)/\ell$ for edges at the minimal spacing $m(n)$, in units of the side $\ell = 2\sin\frac{\pi}{n}$. By Corollary 2.3, with $c = c_{ef}$,

$$\tau(n) = \frac{\operatorname{dist}(c, D)}{\ell}, \qquad |c| = 4a\sin\frac{\pi m}{n}. \tag{2}$$

Theorem 7.1 (Tension)**.** *(i) For $3 \le n \le 6$, $\tau(n) = 0$.*

(ii) For $n \ge 7$: $\tau(n) = 0 \iff 6 \mid n \iff k(n) = 6$. Thus a maximally tense packing is exactly a hexagonal resonance, and this is a theorem rather than an empirical observation.

(iii) Let $n \ge 7$, $6 \nmid n$, $r = n \bmod 6$, $\kappa = \frac{6-r}{6}$ and $u = \frac{\pi}{n}$. Then, for all such n,

$$\Big|\,\tau(n) - \sqrt{3}\,\kappa\,\Big| \;\le\; \Big(\frac{\kappa^2}{2} + 1\Big)\,u.$$

More precisely, with $\Lambda_n := (|c| - 2a)/\ell$,

$$\Lambda_n = \sqrt{3}\,\kappa - \frac{\kappa^2}{2}\,u + \theta_n\,u^2 \;\; (-0.9 \le \theta_n \le 0.26), \qquad 0 \le \Lambda_n - \tau(n) \le \tan\tfrac{u}{2} \le 0.51\,u.$$

In particular $\tau(n) \to \sqrt{3}\,\kappa = \frac{\sqrt{3}}{6}\,\big((-n) \bmod 6\big)$ along each residue class $r \ne 0$, with an error of order $1/n$ and explicit constants (Table 3).

r	κ	$\sqrt{3}\,\kappa$	$(\frac{\kappa^2}{2}+1)\pi$	observed max	ratio
1	5/6	1.4434	4.232	3.104	0.73
2	4/6	1.1547	3.840	1.238	0.32
3	3/6	0.8660	3.534	1.920	0.54
4	2/6	0.5774	3.316	1.954	0.59
5	1/6	0.2887	3.185	1.315	0.41

Table 3: For $r = n \bmod 6 \in \{1,\dots,5\}$ and $\kappa = \frac{6-r}{6}$: the limit $\sqrt{3}\kappa$ of the tension, the constant $(\frac{\kappa^2}{2}+1)\pi$ in Theorem 7.1(iii) (the bound reads $|\tau(n) - \sqrt{3}\kappa| \le (\frac{\kappa^2}{2}+1)\frac{\pi}{n}$), the largest value of $n\,|\tau(n) - \sqrt{3}\kappa|$ found for $7 \le n \le 3000$ ("observed max"), and the ratio of the last two.

Proof. (i) For $n \le 6$, $m = 1$: the neighbours on the adjacent edges e and $e+1$ both contain the common vertex v_{e+1} of those edges (it is an endpoint of edge $e \subseteq N_e$ and of edge $e+1 \subseteq N_{e+1}$), so their distance is 0.

(ii) If $6 \mid n$, then $\tau = 0$ by Theorem 3.2(c) (they touch). If $6 \nmid n$, then $m = d_0$ in the proof of Theorem 3.2(b), where $|c| > R_D$ was shown; hence $c \notin D$, $\operatorname{dist}(c, D) > 0$ and $\tau > 0$. The equivalence with $k = 6$ is Proposition 4.3.

(iii) Let $n \ge 7$, $6 \nmid n$. Then $u \le \pi/7 = 0.4488\ldots$ and $\kappa \le \frac{5}{6}$. We prove the three displayed statements in four steps.

Step 1: τ is within $\tan\frac{u}{2}$ below Λ_n. Since $D \subseteq \{|x| \le R_D\}$ we have $\operatorname{dist}(c, D) \ge |c| - R_D$. Since $|c| > R_D \ge r_D$, the point $\frac{r_D}{|c|}c$ lies in the disc of radius r_D, hence in D, so $\operatorname{dist}(c, D) \le |c| - r_D$. Dividing by ℓ and subtracting $\Lambda_n = (|c| - 2a)/\ell$,

$$\frac{2a - R_D}{\ell} \;\le\; \tau(n) - \Lambda_n \;\le\; \frac{2a - r_D}{\ell}.$$

By Lemma 2.4: for even n, $R_D = 2$, $r_D = 2a$; for odd n, $R_D = 2\cos\frac{u}{2} \le 2$ and $r_D = 1 + a \ge 2a$. In both cases $2a - r_D \le 0$ and $2a - R_D \ge 2a - 2$, so

$$-\frac{2 - 2a}{\ell} \le \tau(n) - \Lambda_n \le 0, \qquad \frac{2 - 2\cos u}{\ell} = \frac{2(1 - \cos u)}{2\sin u} = \tan\frac{u}{2}.$$

For the last inequality in the theorem: $\tan y / y$ is increasing on $(0, \frac{\pi}{2})$, so for $y = \frac{u}{2} \le \frac{\pi}{14}$ we get $\tan\frac{u}{2} \le \frac{u}{2} \cdot \frac{\tan(\pi/14)}{\pi/14} = 1.0172 \cdot \frac{u}{2} \le 0.51\,u$.

Step 2: closed form of Λ_n. Write $n = 6q + r$. Then $m = q + 1$ and $\frac{m}{n} - \frac{1}{6} = \frac{6(q+1) - n}{6n} = \frac{6 - r}{6n}$, so $\frac{\pi m}{n} = \frac{\pi}{6} + x$ with $x := \kappa u$ (and $0 < x \le \frac{5}{6} \cdot \frac{\pi}{7} = 0.374$). Using $\sin(\frac{\pi}{6} + x) = \frac{1}{2}\cos x + \frac{\sqrt{3}}{2}\sin x$ in (2),

$$|c| = 4a\sin\left(\tfrac{\pi}{6} + x\right) = 2a\,(\cos x + \sqrt{3}\sin x),$$
$$\Lambda_n = \frac{|c| - 2a}{2\sin u} = \cos u \cdot \frac{N(x)}{\sin u}, \qquad N(x) := \cos x + \sqrt{3}\sin x - 1.$$

Note $N(x) = 2\sin(x + \frac{\pi}{6}) - 1 \ge 0$ for $x \ge 0$.

Step 3: four elementary inequalities. For $y \ge 0$,

$$\text{(E1)}\ \sin y \le y, \quad \text{(E2)}\ \cos y \ge 1 - \tfrac{y^2}{2}, \quad \text{(E3)}\ \sin y \ge y - \tfrac{y^3}{6}, \quad \text{(E4)}\ \cos y \le 1 - \tfrac{y^2}{2} + \tfrac{y^4}{24}.$$

They follow by successive integration: (E1) integrates $\cos t \le 1$ over $[0, y]$; (E2) integrates (E1) via $\cos y = 1 - \int_0^y \sin t\,dt$; (E3) integrates (E2) via $\sin y = \int_0^y \cos t\,dt$; (E4) integrates (E3) via $\cos y = 1 - \int_0^y \sin t\,dt$. With $x = \kappa u$ and (E1)–(E4) applied to N:

$$N_- := \sqrt{3}\kappa u - \tfrac{\kappa^2 u^2}{2} - \tfrac{\sqrt{3}}{6}\kappa^3 u^3 \;\le\; N(x) \;\le\; N_+ := \sqrt{3}\kappa u - \tfrac{\kappa^2 u^2}{2} + \tfrac{\kappa^4 u^4}{24},$$

and, for the other factors, $u - \frac{u^3}{6} \le \sin u \le u$ and $1 - \frac{u^2}{2} \le \cos u \le 1$. Put $L := \sqrt{3}\kappa - \frac{\kappa^2}{2}u$.

Step 4a: upper bound for Λ_n. Since $\cos u \le 1$ and $N \ge 0$,

$$\Lambda_n \le \frac{N_+}{\sin u} \le \frac{N_+}{u(1 - u^2/6)} = \frac{A}{1 - t}, \qquad A := \sqrt{3}\kappa - \tfrac{\kappa^2}{2}u + \tfrac{\kappa^4}{24}u^3, \quad t := \tfrac{u^2}{6} \le 0.0336.$$

Now $\frac{1}{1-t} = 1 + \frac{t}{1-t} \le 1 + 1.035\,t$, and $A \le \sqrt{3} \cdot \frac{5}{6} + \frac{(5/6)^4}{24}\left(\frac{\pi}{7}\right)^3 \le 1.445$. Hence

$$\Lambda_n - L \le (A - L) + 1.035\,t\,A \le \tfrac{\kappa^4}{24}u^3 + 1.035 \cdot \tfrac{u^2}{6} \cdot 1.445 \le u^2\left(\tfrac{(5/6)^4}{24} \cdot 0.4489 + 0.2492\right) \le 0.26\,u^2.$$

Step 4b: lower bound for Λ_n. Since $\sin u \le u$, $N \ge N_- \ge 0$ (indeed $N_- = u\kappa\big(\sqrt{3} - \frac{\kappa u}{2} - \frac{\sqrt{3}}{6}\kappa^2 u^2\big) > 0$ for $\kappa u \le 0.374$) and $\cos u \ge 1 - \frac{u^2}{2} > 0$,

$$\Lambda_n \ge \left(1 - \frac{u^2}{2}\right)\frac{N_-}{u} = \left(1 - \frac{u^2}{2}\right)\left(L - \tfrac{\sqrt{3}}{6}\kappa^3 u^2\right) \;\ge\; L - \tfrac{\sqrt{3}}{6}\kappa^3 u^2 - \tfrac{u^2}{2}\sqrt{3}\kappa,$$

where in the last step we used $0 \le L - \frac{\sqrt{3}}{6}\kappa^3 u^2 \le \sqrt{3}\kappa$. As $\kappa \le \frac{5}{6}$, $\frac{\sqrt{3}}{6}\kappa^3 + \frac{\sqrt{3}}{2}\kappa \le 0.1671 + 0.7217 = 0.8888 \le 0.9$, so $\Lambda_n - L \ge -0.9\,u^2$. Steps 4a and 4b prove $\Lambda_n = \sqrt{3}\kappa - \frac{\kappa^2}{2}u + \theta_n u^2$ with $-0.9 \le \theta_n \le 0.26$.

Step 5: combination. By Step 1 and Steps 4a, 4b,

$$\begin{aligned}|\tau - \sqrt{3}\kappa| &\le |\tau - \Lambda_n| + |\Lambda_n - L| + |L - \sqrt{3}\kappa| \\ &\le 0.51\,u + 0.9\,u^2 + \tfrac{\kappa^2}{2}u \;\le\; \tfrac{\kappa^2}{2}u + u\,(0.51 + 0.9 \cdot 0.4489) \;\le\; \left(\tfrac{\kappa^2}{2} + 1\right)u,\end{aligned}$$

since $0.51 + 0.404 = 0.914 < 1$. □

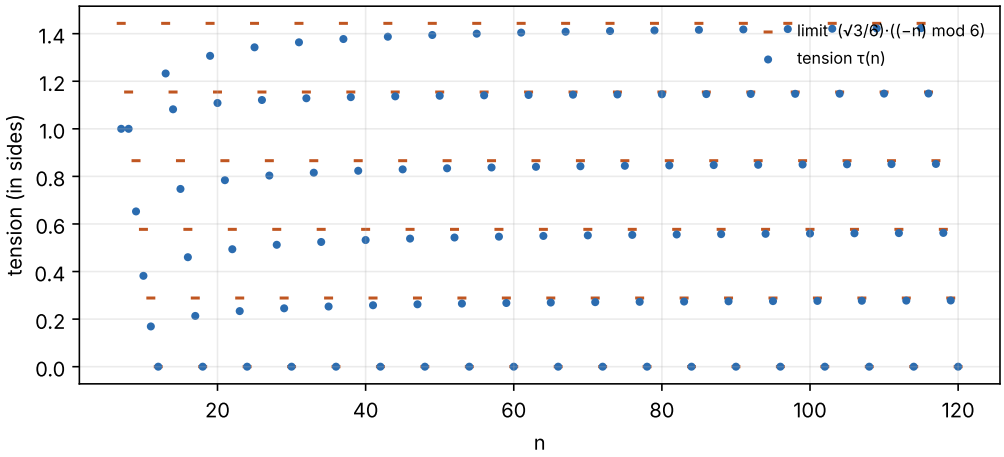


Figure 2: The exact tension $\tau(n)$ (dots) for $7 \le n \le 120$ and its limiting sawtooth (dashes; period 6, zeros at the resonances).

Proposition 7.2 (Discretised kissing number)**.** *Let $\phi(n) = k(n)\,m(n)/n$ be the fraction of edges occupied by an extremal placement. Then $\limsup k(n) = 6$ and $\liminf k(n) = 5$, so the conjoinability has no limit; $\phi(n) = 1$ for $6 \mid n$, while $\phi(n) \to \frac{5}{6}$ along every other residue class. Equivalently, the circle's kissing configuration (neighbours at angular spacing $\pi/3$) is attained by the polygon exactly when $\pi/3 \in \frac{2\pi}{n}\mathbb{Z}$, i.e. $6 \mid n$.*

Proof. By Proposition 4.3, $k = 5 + [6 \mid n]$ for $n \ge 20$, which gives $\limsup = 6$, $\liminf = 5$. For $6 \nmid n$, $m = \frac{n+6-r}{6}$ and $\phi(n) = \frac{5m}{n} = \frac{5}{6}\big(1 + \frac{6-r}{n}\big) \to \frac{5}{6}$; for $6 \mid n$, $\phi = \frac{6 \cdot n/6}{n} = 1$. Last statement: $\frac{\pi}{3} = \frac{2\pi}{n} \cdot \frac{n}{6}$. □

Remark 7.3 (What this does to the hand-assigned labels)*.* In the spreadsheet the label "Maximally Tense" occurs at $n = 13, 18, 24, 30, 36, 42, 48, 54$, but not at $n = 6, 12$ or 174 (all with $k = 6$), and $n = 13$ has $k = 4$. So the labels are not a function of the conjoinability, and "all $k = 6$ points are Maximally Tense" is false as a statement about the labels (it holds for $18 \le n \le 54$ only). Theorem 7.1(ii) replaces it by a measurable quantity for which it is true for every n. Rigidity (Theorem 5.3 and Corollary 5.4) is the objective version of "resonance collapses the isomers to one perfect configuration": the extremal placement is unique up to rotation iff $\sigma \le 1$.

8 Status of the hypotheses

Hypothesis	Formal statement	Status
H1 resonance of the conjoinability	$k(n) = \lfloor n/\lceil n/6 \rceil \rfloor$; $k(n) = 6 \iff 6 \mid n$ (Thm. 4.1, Prop. 4.3)	Proved (Thm. 3.2)
H2 vertex skipping	$s(n) = \max(0, \lceil n/6 \rceil - 2)$, non-decreasing; explicit values of k, σ, s (Thm. 4.2)	Proved
H3 generating function	$P_n(x) = \sum \lambda_i^n$, $\lambda^m = \lambda^{m-1} + x$; $\deg P_n = 6 \iff 6 \mid n$ (Thm. 6.1, Cor. 6.2)	Proved
H4 tension	$\tau(n) = 0 \iff 6 \mid n$ (Thm. 7.1(ii)); the label-based version is false as stated	Replaced by a proved statement
H5 discrete-to-continuous	sawtooth limit of τ with the explicit bound of Thm. 7.1(iii); Prop. 7.2	Proved. Convergence of the full overlap profile to the disc's lens profile is *not* proved here
H6 flexibility and rigidity	counts (Thm. 5.1), rigid n (Thm. 5.3, Cor. 5.4), tight chains (Thm. 5.5)	Proved

Not covered. The only external ingredient is the convex-analysis identity $\operatorname{ri}(C_1+C_2) = \operatorname{ri} C_1 + \operatorname{ri} C_2$ (cited in Lemma 2.2); the numerical values quoted in Lemma 3.1 and Proposition 3.4 are finite checks.

9 Computational cross-checks

The statements above were also checked by program (exact polygon intersection with `shapely`; scripts `Polygon_solver.py` and `Polygon_placements.py`, test suite of 100 tests):

- Lemma 2.1 for $n \in \{7, 8, 13, 25, 34, 36, 41\}$ and all edges; Corollary 2.3 for all $5 \le n \le 80$ and all $d \le n/2$ (1596 cases, no mismatch); Lemma 2.4 for $n \in \{8, 12, 13, 25\}$.
- Theorem 3.2: the exact solver's smallest admissible distance equals $\lceil n/6 \rceil$ for every $n \le 400$, and overlap at *every* distance $d \le n/2$ agrees with "overlap iff $d < n/6$" for every $n \le 160$; the two inequalities of the proof (Lemma 3.1 applied to d_0) were additionally checked for every $n \le 2 \cdot 10^6$.
- Proposition 3.3: for $n = 12, 18, 24, 36, 60, 102$ the neighbours on edges 0 and $n/6$ share exactly one full edge: two vertices and the segment between them, of length ℓ. (The program compares vertices: exactly two coincide, at distance ℓ; since vertices of a regular polygon at distance ℓ are adjacent, the segment between them is an edge of both. Generic polygon-intersection routines return unstable shapes for exactly touching polygons, so the code uses a tolerance of 10^{-12} on the overlap area.)
- Theorems 4.1–4.2 and Corollary 4.4: the identities $\sigma = n \bmod m$ and $\sigma = q + r - 5$ (when $k = 5$) and the lists of n with $\sigma = 0$ and $\sigma = 1$ for all $7 \le n \le 20000$; Theorem 5.3 and Corollary 5.4 by enumeration for $7 \le n \le 400$.
- Theorem 5.1 and Proposition 5.2: brute-force enumeration of all subsets using pairwise polygon intersections for $n \le 16$, and enumeration versus closed form for larger n (including the worked example $R(34, 5) = 14$, $B(34, 5) = 10$).
- Theorem 6.1: coefficient identity for $1 \le m \le 12$, $m \le n < 140$, and a numerical root check.
- Theorem 7.1: the three displayed inequalities of (iii) and the intermediate bounds $-0.9 \le \theta_n \le 0.26$ and $0 \le \Lambda_n - \tau \le \tan \frac{u}{2}$ were checked for every $7 \le n \le 3000$, $6 \nmid n$ (largest observed ratio $|\tau - \sqrt{3}\kappa|/((\kappa^2/2 + 1)u) = 0.73$; Table 3).

Note. The code and first drafts of the proofs were prepared with AI assistance (Claude); the problem, definitions, drawings and data are the author's.